

Homework #1: Vector Algebra; Fields; Lorentz Force; Integration Practice

4. Given

$$\mathbf{A} = 3\mathbf{a}_x + x\mathbf{a}_y + y\mathbf{a}_z$$

$$\mathbf{B} = x^2\mathbf{a}_x + 4\mathbf{a}_y$$

- (a) Determine $\mathbf{A} \cdot \mathbf{B}$ at the point $x = 2, y = 3$.
- (b) Find the angle between $\mathbf{A}$ and $\mathbf{B}$.
- (c) Find the projection of $\mathbf{B}$ along the direction of $\mathbf{A}$.

5. Given the vectors

$$\mathbf{A} = 5\mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z$$

$$\mathbf{B} = B_x\mathbf{a}_x + 2\mathbf{a}_y + B_z\mathbf{a}_z$$

$$\mathbf{C} = 3\mathbf{a}_x + C_y\mathbf{a}_y + \mathbf{a}_z$$

Find B_x , B_z , and C_y so that $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are mutually orthogonal (i.e., perpendicular to each other).

6. Given two vectors, $\mathbf{B} = 2\mathbf{a}_x + \mathbf{a}_y + 3\mathbf{a}_z$ and $\mathbf{C} = 2\mathbf{a}_y + 6\mathbf{a}_z$

- (a) Find a unit vector perpendicular to the plane containing $\mathbf{B}$ and $\mathbf{C}$. Draw a schematic diagram illustrating your answer.
- (b) Find the area of the parallelogram constructed from $\mathbf{B}$ and $\mathbf{C}$.

7. Two vectors $\mathbf{A}$ and $\mathbf{B}$ have the same origin and are given by

$$\mathbf{A} = \mathbf{a}_\rho + \mathbf{a}_\phi + 3\mathbf{a}_z$$

$$\mathbf{B} = \alpha\mathbf{a}_\rho + \beta\mathbf{a}_\phi - 6\mathbf{a}_z$$

- (a) Determine α and β such that the two vectors are parallel.
- (b) Express $\mathbf{A}$ in the Cartesian coordinate system.

8. Given

$$\mathbf{A} = \cos \phi \mathbf{a}_\rho + \sin \phi \mathbf{a}_\phi + \rho \mathbf{a}_z$$

$$\mathbf{B} = \rho \mathbf{a}_\rho + \phi \mathbf{a}_\phi + 2\mathbf{a}_z$$

with ϕ expressed in radians. Determine $\mathbf{A} \cdot \mathbf{B}$ at the point $x = 2, y = 3$. Assume that the origins of the cylindrical and Cartesian coordinates are the same, and the x axis coincides with $\phi = 0$. Express the vector $\mathbf{A}$ in Cartesian coordinates.

23. A charged particle (charge q , mass m) is moving under the influence of a magnetic field

$$\mathbf{B} = B_0 \mathbf{a}_z.$$

If the velocity vector of the particle is

$$\mathbf{v} = v_x \mathbf{a}_x + v_y \mathbf{a}_y + v_z \mathbf{a}_z$$

Show that the acceleration of this particle lies in the x - y plane.

24. A magnetic field $\mathbf{B} = B_0(\mathbf{a}_x + 2\mathbf{a}_y - 4\mathbf{a}_z)$ exists at a point. What should be the electric field at that point if the force experienced by a test charge moving with a velocity $\mathbf{v} = v_0(3\mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z)$ is to be zero?

26. Consider the contour c shown in Figure P1.26 and the vector field

$$\mathbf{F} = 2\rho(z^2 + 1) \cos \phi \mathbf{a}_\rho - \rho(z^2 + 1) \sin \phi \mathbf{a}_\phi + 2\rho^2 z \cos \phi \mathbf{a}_z.$$

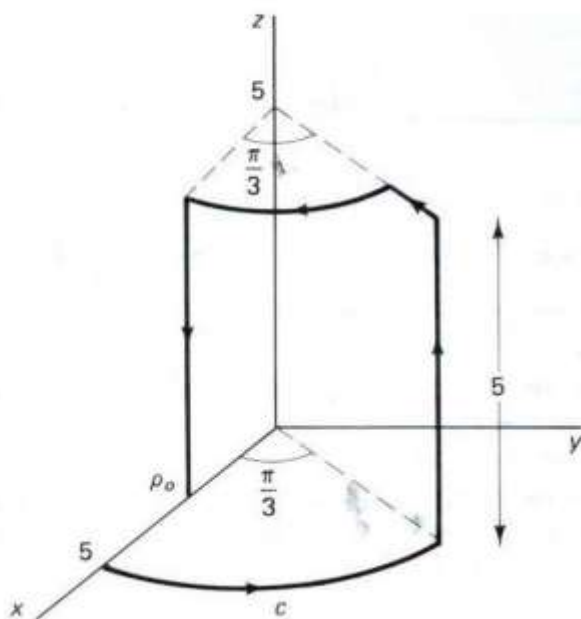


Figure P1.26

- Evaluate $\int_c \mathbf{F} \cdot d\boldsymbol{\ell}$.
- Evaluate $\int_{c_1} \mathbf{F} \cdot d\boldsymbol{\ell}$ where c_1 is the straight line joining $(\rho = \rho_0, \phi = 0, z = 0)$ to $(\rho = 5, \phi = 0, z = 0)$.
- Are the results of *a* and *b* consistent with the field $\mathbf{F}$ being conservative? (A field is conservative when its line integral along any closed contour is zero.)